
HARARI PEOPLE REGIONAL STATE · ETHIOPIA

Trade, Industry & Tourism Development Bureau

Harari PCC Portal

Professional Competence Certificate Platform

A Proposal for Official Adoption and Regional Deployment
of a Digital Business Licensing Portal for the Harari Region

Submitted to: Harari Regional State Council

Submitted by: Trade, Industry & Tourism Development Bureau

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1. Executive Summary

The Harari People Regional State, despite its cultural and commercial significance as the home of Harar Jugol — a UNESCO World Heritage Site — continues to rely on a paper-based, in-person business licensing process that is slow, opaque, and difficult to audit. Citizens travel to the Bureau office in Harar, queue for hours, submit paper documents by hand, and wait an average of five to fourteen business days for a decision. There is no public registry to verify issued certificates, leaving room for forgery and undermining trust in the regional licensing system.

This proposal presents the Harari PCC Portal — a complete, working digital platform that moves the entire Professional Competence Certificate (PCC) workflow online. Citizens register, submit documents, complete a competence assessment, and receive a verifiable certificate without visiting a physical office. Regional officers review applications through a dedicated console with full audit logging. The platform is built on modern, secure, scalable technology and is already functional as a working prototype.

We request the Regional Council's formal adoption of this platform as the official digital channel for PCC issuance, along with a budget allocation of ETB 2,460,000 for production deployment, capacity building, and the first twelve months of operation. The expected outcomes are: a reduction in average processing time from 5–14 days to under 24 hours; full visibility of application status for every citizen; and a public certificate verification endpoint that eliminates forgery risk. The platform aligns with federal digital transformation priorities and with the Harari Region's broader goal of becoming a model small-region administration in Ethiopia.

Key Ask: Formal adoption of the Harari PCC Portal as the official digital PCC channel, plus a twelve-month implementation and operating budget of ETB 2,460,000.

2. Background and Problem Analysis

2.1 Current State of Business Licensing in the Harari Region

Under the existing arrangement, any citizen of the Harari Region who wishes to open a business must obtain a Professional Competence Certificate (PCC) from the Trade, Industry & Tourism Development Bureau before applying for a trade licence. The PCC process is designed to verify that the applicant understands basic Ethiopian business law, taxation, labour regulations, and regional requirements, and that they hold the educational or professional qualifications appropriate to their intended business sector.

In practice, every step of this process is paper-based. The applicant obtains a paper form from the Bureau office in Harar, completes it by hand, attaches photocopies of

supporting documents (national ID, educational certificates, business plan, lease agreement), and submits the folder in person. The folder is then routed manually between the front desk, a reviewing officer, and the head of the licensing division. The applicant has no way to know where their file is in the queue. When the decision is ready, the applicant must return to the Bureau to collect the certificate in person.

2.2 Specific Pain Points

- Processing time of 5–14 business days per application, with peaks of three weeks during high-demand periods such as the run-up to trade-fair season.
- No status visibility — applicants cannot track their application and must call or visit the Bureau to enquire, consuming staff time on both sides.
- No public verification — third parties (banks, landlords, partner businesses) cannot validate an applicant's certificate without phoning the Bureau, creating forgery risk and friction in commercial transactions.
- Lost or misrouted files — paper folders are occasionally misplaced during internal routing, forcing applicants to resubmit documents and inflating the actual processing time further.
- Limited service reach — citizens living in woredas outside Harar must travel to the regional capital for every step of the process, imposing a hidden cost on rural entrepreneurs.
- No management analytics — the Bureau leadership has no real-time view of pending applications, sectoral distribution, or reviewer workload, making resource planning reactive.

2.3 Policy and Strategic Context

The federal government of Ethiopia has, since 2022, made digital transformation a national priority under the "Digital Ethiopia 2025" strategy. Several regional states — including Oromia, Amhara, and the Sidama Region — have begun deploying online portals for selected trade and land services. The Harari Region, despite being the smallest regional state by area, has lagged behind in this regard, in part because no tailored solution existed that matched the Region's specific Bureau workflow and cultural context.

Adopting the Harari PCC Portal directly addresses this gap. It is built specifically for the Harari Region — not a generic off-the-shelf product — and aligns with the regional Bureau's actual decision-making workflow, document requirements, and cultural identity.

3. Project Objectives and Expected Outcomes

3.1 Primary Objectives

The project has four primary objectives, each tied to a measurable outcome:

- Digitise the end-to-end PCC workflow — from registration to certificate issuance — so that no physical office visit is required for any step of an application.
- Enforce competence verification through a structured 17-question assessment covering Ethiopian business law, taxation, labour law, consumer protection, and Harari regional regulations.
- Provide a public, no-authentication certificate verification endpoint so that any third party can validate an issued certificate by its unique number.
- Give Bureau leadership a real-time dashboard of application volume, status distribution, sectoral breakdown, and reviewer activity for operational decision-making.

3.2 Expected Outcomes and Indicators

Outcome	Indicator	Baseline	Target (12 mo)
Faster processing	Average time from submission to decision	5–14 days	Under 24 hours
Citizen transparency	Share of applicants who can check status online	0%	100%
Fraud reduction	Public verification endpoint availability	None	Live, no-auth endpoint
Geographic reach	Applications submitted from outside Harar city	Unknown	Tracked and reported quarterly
Operational visibility	Bureau dashboard with live queue + analytics	None	Deployed and used daily

Table 1. Expected outcomes and measurable indicators.

4. Solution Description

4.1 Platform Overview

The Harari PCC Portal is a web-based platform accessible from any modern browser on desktop or mobile. It serves three user roles with strictly enforced, server-side access control: applicants, reviewers (Bureau officers), and administrators. Each role sees only the functionality and data appropriate to it.

The platform implements a five-step application wizard that guides the applicant through personal information, business information, document upload, competence assessment, and a final review-and-submit step. A progress indicator at the top of the wizard shows the applicant exactly where they are in the process. Drafts are saved automatically, so an applicant can pause at any step and return later without losing their work.

4.2 Key Functional Capabilities

For Applicants

- Self-service registration with national ID, contact information, and regional address details.
- Five-step guided application wizard with autosave and progress tracking.
- Document upload supporting 19 document types aligned to the Ethiopian education system, including Grade 8, Grade 10 (EGECE), Grade 12 (matric), TVET certificates, diplomas, advanced diplomas, bachelor's, master's, and doctoral degrees, plus professional certifications and supporting business documents.
- Randomised 10-question competence assessment drawn from a 17-question bank, with a 70% pass mark and unlimited retries. Each incorrect answer is followed by an explanation so the applicant learns from every attempt.
- Real-time status tracking (Draft → Submitted → Under Review → Approved / Rejected → Certificate Issued) with a full audit-trail timeline visible per application.
- One-click certificate viewing and printing, with the option to save the certificate as a PDF.

For Reviewers and Bureau Officers

- Application queue with full-text search and status filters (Submitted, Under Review, Issued, Rejected).
- One-click workflow: Claim → Approve (auto-issues certificate) or Reject (with mandatory written reason).
- Educational qualifications summary card per applicant, displaying all verified credentials at a glance.
- Dashboard analytics with counts by status, certificates issued by sector, and a live recent-activity feed.
- Full audit log — every action (login, claim, approve, reject, issue) is recorded with user, timestamp, and IP address, supporting accountability and after-the-fact investigation.

For the Public

- A public certificate verification endpoint. Any third party — a bank, a landlord, a partner business, a customer — can validate a certificate by entering its unique number (e.g. HRS-PCC-CERT-2026-0001) and receiving an instant valid/invalid response with the certificate holder's name, business name, and validity period.

4.3 Cultural and Regional Alignment

Unlike generic licensing software, the Harari PCC Portal is designed to feel native to the Region. The visual identity draws on Harari cultural motifs — the eight-pointed star, the Harar Jugol gate, and a colour palette of royal purple, gold, terracotta, cream, and Islamic green inspired by the historic walled city. The institutional emblem combines traditional Ethiopian symbolic elements (balance scale for justice, sun for hope, castle for heritage, gear for industry) reimagined in the regional palette. This cultural rooting matters: citizens are more likely to trust and adopt a service that visibly belongs to their own Region rather than a generic national template.

5. Technical Architecture

5.1 Technology Stack

The platform is built on a modern, industry-standard technology stack that is mature, well-documented, and widely supported by Ethiopian and international developers. This ensures the Region is not locked into a single vendor and can hire local talent to maintain and extend the system.

Layer	Technology	Role
Frontend	Next.js 16 + TypeScript + Tailwind CSS + shadcn/ui	Responsive web interface, mobile-first design
Backend	Next.js API routes (Node.js 20+)	REST API with 12 endpoint groups
Database	Prisma ORM + PostgreSQL (production)	Relational storage with type-safe access
Authentication	bcryptjs password hashing + JWT sessions in httpOnly cookies	Secure, stateless, scalable auth
File storage	S3-compatible object storage (e.g., MinIO, AWS S3)	Encrypted document storage with signed URLs
Security	Role-based access control enforced server-side	Applicant / Reviewer / Admin permissions

Table 2. Production technology stack.

5.2 Data Model and Security

The platform stores seven core entities: user profiles, applications, documents, certificates, audit logs, notifications, and a sequence counter for reference-number generation. All sensitive fields — passwords, document files, audit logs — are protected at the application layer (bcrypt hashing, signed-URL access, server-side role checks) and at the database layer (PostgreSQL row-level security policies can be enabled for defence in depth).

Document files (PDFs, images of certificates, lease agreements) are stored in an S3-compatible object store rather than in the database itself, which keeps the database small and fast and allows individual documents to be expired, backed up, or migrated independently. Access to documents is mediated by short-lived signed URLs that expire within sixty seconds, so even if a URL is leaked it cannot be reused.

6. Implementation Plan

6.1 Phased Approach

We propose a four-phase implementation over twelve months. Each phase concludes with a concrete deliverable that the Bureau can review and approve before the next phase begins.

Phase	Duration	Key Activities	Deliverable
1. Production deployment	Months 1–2	Provision production infrastructure (database, storage, hosting). Migrate from SQLite to PostgreSQL. Configure backups, monitoring, and a custom domain (pcc.harariregion.gov.et).	Live production environment
2. Pilot rollout	Months 3–4	Onboard the first cohort of 50 applicants and 3 reviewers from the Bureau. Run parallel	Pilot evaluation report

		paper + digital processing. Collect feedback weekly.	
3. Full rollout & training	Months 5–8	Train all Bureau licensing staff. Open the portal to the general public. Decommission paper-only intake. Run public awareness campaign.	Public launch + trained staff
4. Stabilisation & handover	Months 9–12	Bug fixes, performance tuning, first analytics review. Hand over operational responsibility to the Bureau IT unit. Prepare roadmap for year-2 enhancements.	Operational handover

Table 3. Four-phase implementation plan over twelve months.

6.2 Key Milestones

- Month 2: Production environment live at pcc.harariregion.gov.et.
- Month 4: Pilot evaluation report delivered to the Bureau leadership.
- Month 5: All Bureau licensing staff trained and certified on the platform.
- Month 6: Public launch; paper-only intake decommissioned.
- Month 12: Operational handover to the Bureau IT unit; year-2 roadmap agreed.

7. Budget

The total requested budget for the twelve-month implementation and first-year operation is ETB 2,460,000. The budget is divided into four categories: one-time setup, software development and customisation, capacity building and training, and recurring operating costs. All figures are in Ethiopian Birr and are estimates based on typical Ethiopian IT project costing; final figures will be confirmed during procurement.

Category	Line item	ETB
One-time setup	Domain registration (pcc.harariregion.gov.et) — 3 years	15,000

	SSL certificate — 3 years	30,000
	Production hosting (cloud VPS, 4 vCPU / 8 GB) — 12 mo	180,000
	PostgreSQL managed database — 12 mo	144,000
	S3-compatible object storage (50 GB) — 12 mo	60,000
	Backup and disaster recovery setup	75,000
Development	Production hardening, security audit, PostgreSQL migration	350,000
	Integration with ERCA TIN verification API	180,000
	Amharic / Arabic localisation (UI + assessment)	120,000
	Mobile-responsive optimisation and PWA support	150,000
	Bug fixing and feature polishing (months 9–12)	200,000
Capacity building	Training of 10 Bureau staff (3 days @ ETB 8,000/day)	240,000
	Training materials, manuals, video tutorials	60,000
	Public awareness campaign (radio, social, flyers)	150,000
Operations	System administrator (part-time, 12 mo)	300,000
	Monitoring, security patching, incident response	120,000
	Contingency (10%)	186,000
	TOTAL	2,460,000

Table 4. Detailed project budget in Ethiopian Birr (ETB).

8. Risk Analysis and Mitigation

We have identified five principal risks to successful delivery and adoption. None of them is unusual for a government digital-service project, and each has a concrete mitigation plan.

Risk	Likelihood	Impact	Mitigation
Low digital literacy among rural applicants	Medium	Medium	Provide a printable step-by-step guide in Amharic; offer assisted-application windows at the Bureau office during the first six months.
Staff resistance to new workflow	Medium	High	Involve Bureau staff in pilot feedback; run three-day hands-on training; designate a staff champion in each division.
Internet connectivity outages	Medium	Medium	The portal is designed to gracefully resume in-progress applications; offline data entry can be done at the Bureau office using the staff console.
Data breach or document leakage	Low	High	All documents stored in encrypted object storage; access mediated by short-lived signed URLs; full audit log of every access; quarterly security review.
Scope creep (new feature requests during build)	High	Medium	Strict change-control process: any new feature requested after Month 2 is logged and scheduled for Year 2

			unless it blocks launch.
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Table 5. Risk register with mitigation actions.

9. Sustainability

Beyond the first twelve months, the platform is designed to be sustainable in three respects: technical, financial, and institutional.

9.1 Technical Sustainability

The technology stack is built on widely-used open-source frameworks (Next.js, PostgreSQL, Prisma). This means the Bureau is not dependent on any single vendor for maintenance or enhancements — any competent Ethiopian software developer can be onboarded to support the system. The codebase is TypeScript throughout, which reduces the cost of future changes by catching errors at compile time. Documentation — including a comprehensive README, API reference, and operational runbook — will be delivered as part of the handover.

9.2 Financial Sustainability

Recurring annual operating costs after Year 1 are estimated at ETB 420,000 (hosting, database, storage, domain, SSL, part-time system administrator). These costs can be absorbed into the Bureau's regular IT operating budget. The Region may also, in future, introduce a small digital-service fee per application (as several other regions have done) to fully recover operating costs from applicants who choose to use the digital channel.

9.3 Institutional Sustainability

Operational responsibility for the platform will be transferred to the Bureau IT unit by Month 12. The handover will include: source code, documentation, operational runbooks, monitoring dashboards, and a one-month shadow-support period during which the development team remains available for escalation. The Bureau IT unit will be trained to handle routine operations (user management, application review support, certificate verification, backup verification) without external assistance.

10. Conclusion and Requested Action

The Harari PCC Portal is a complete, working, region-specific digital platform that addresses a real, documented pain point in the daily experience of Harari entrepreneurs. It is not a concept or a mockup — it is a functioning prototype that can be deployed to production within two months of approval. It aligns with federal digital-transformation

priorities, respects the Region's cultural identity, and is built on a maintainable open-source stack.

We respectfully request the Regional Council to:

- Formally adopt the Harari PCC Portal as the official digital channel for Professional Competence Certificate issuance in the Harari Region.
- Approve a twelve-month implementation and operating budget of ETB 2,460,000, drawn from the Bureau's innovation and digital-transformation allocation.
- Designate a project sponsor at the level of Bureau Head, with authority to coordinate across IT, licensing, and communications functions.
- Authorize a public launch by Month 6 of the implementation timeline, accompanied by the planned awareness campaign.

We are available at the Council's convenience to present the working prototype in a live demonstration, to answer technical or operational questions, and to refine the proposal based on the Council's guidance.

Submitted by:

Trade, Industry & Tourism Development Bureau

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